

Proceedings of the 40th National Conference on Mechanical Engineering of CSME: Taiwanese Sign Language Visual Recognition

Yu-Chen Chen¹, Hang-Pan Huang^{1*}, Yun-Chieh Lee¹, Jin-Rou Chang¹, Diego Marecos¹

¹Department of Mechanical Engineering, National Taiwan University, Taipei, Taiwan.

(MOST110-2221-E-002-110-MY3)

*Email: hanpang@ntu.edu.tw (Corresponding author)

Abstract

This paper presents two visual recognition approaches for Taiwanese Sign Language (TSL) aimed at recognizing TSL vocabulary. One involves breaking down TSL into handshape, orientation, position, movement, and facial expressions for separate element recognition. The other involves a Long Short-Term Memory (LSTM) network to establish a recognition model for TSL. Both methods utilize MediaPipe Holistic to extract key point positions of the hands, upper body, and face. Both methods focus on recognizing static TSL vocabularies, and some common vocabularies are used for model testing. Finally, the recognition performance of both methods is compared.

Keywords: Sign Language, Visual Recognition, MediaPipe, LSTM

1. Introduction

In the domain of human-computer interaction, vocal commands are commonly used for communication. However, this poses a challenge for deaf individuals who cannot communicate with robots through speech. To broaden the application of robots and make them more inclusive, it is essential to enable robots to communicate through sign language or to serve as translation tools, translating speech into sign language for deaf individuals and recognizing sign language from regular individuals to produce corresponding speech or text. The realization of a sign language robot involves two key aspects: enabling robots to perform sign language and enabling them to understand the meaning behind sign language gestures. This article will focus on the implementation of the latter, aiming to understand the meanings of TSL gestures through visual recognition.

A sign language sentence can be understood as the natural language syntax that combines a set of sign language vocabularies to convey the meaning. However, since the distinction between sign language vocabularies is typically inferred through experience, most existing research has primarily focused on word recognition in sign language. Therefore, this paper proposes two visual processing and recognition methods for TSL vocabularies, aiming to extract sign language movements from camera footage and analyze sign language gestures and body posture in each frame.

The first method involves breaking down sign language words into their constituent elements. A sign language gesture can be seen as a composition of handshape, orientation, location, movement, and non-manual signals [1]. By recognizing these five elements, we can determine the meaning represented by this combination. We refer to this method as “Sign Language Element Recognition,” and Sections 2.1 to 2.4 will introduce the recognition methods developed for these basic elements. However, as the first method is less flexible and cannot handle dynamic movements, we propose a second recognition approach using a machine learning (ML) framework. Section 2.5 will provide details about the model we employ. In Section 3, we will present and compare the results of both sign language recognition methods.

2. Methods

In order to recognize sign language movements, it is essential to simultaneously detect hands, faces, and upper limbs. Therefore, a human body skeleton detection model capable of recognizing the entire upper body is required. In this study, we utilized the human skeleton detection model, MediaPipe, developed by Google Research in 2020 [2]. MediaPipe is an open-source media ML model application framework. It offers multiple human body detection methods, including hand tracking, hand skeleton detection, facial keypoint detection, etc. We utilized the Holistic model provided by MediaPipe, which enables the recognition of keypoints for the hands, facial mesh, and the entire body. By combining the results of our TSL recognition algorithm with this model, we can merge the recognition results of individual elements or apply ML techniques to deduce the meaning of the sign language gesture.

2.1 Handshape Recognition

Firstly, we established a handshape library for 50 handshapes in TSL, encompassing each handshape's average finger joint angles. Upon detecting the presence of hands in the frame, the system proceeds to calculate the finger joint angles and subsequently compares them to the data within the handshape library. The handshape with the smallest error is considered the result of handshape recognition. The minimum error formula is defined by

$$idx = ref = \min \left\{ \sum_{n=1}^N (\theta_{i_n} - \theta_{ref_n})^2 \right\}$$

$$= \min \left\{ \begin{array}{l} \sum_{n=1}^N (\theta_{i_n} - \theta_{shape_1_n})^2 \\ \sum_{n=1}^N (\theta_{i_n} - \theta_{shape_2_n})^2 \\ \sum_{n=1}^N (\theta_{i_n} - \theta_{shape_k_n})^2 \end{array} \right\} \quad (1)$$

idx corresponds to the reference handshape with the minimum total angle deviation. Here, i represents each individual hand, N is the number of various finger angles to be compared, and we employed a comparison involving 400 angles. Then, θ_{i_n} are all the angles in one hand, while θ_{ref_n} are all the angles in a specific reference handshape.

2.2 Orientation Recognition

We further classify handshapes into two categories: “pointing handshapes” and “palm handshapes (non-pointing handshapes),” based on variations in pointing directions, which involve either finger direction or palm orientation. If a handshape falls into the “pointing handshape” category, we calculate the orientation vector using the key finger. For “palm handshapes,” we determine the vector by computing the normal vector of the palm. Pointing handshapes can be further divided into “single pointing handshapes” and “composite pointing handshapes,” depending on the number of pointing fingers. In the case of “Single pointing handshapes” with only one key finger, we calculate the orientation vector using the fingertip. For “Composite pointing handshapes,” the vectors of multiple fingertips are summed to determine the orientation vector. Once the key vector is determined, we use its Euler angles to establish the orientation of the gesture in space. As illustrated in Fig. 1, in this article, we divide the three-dimensional space into six regions, including upper, lower, left, right, front, and back.

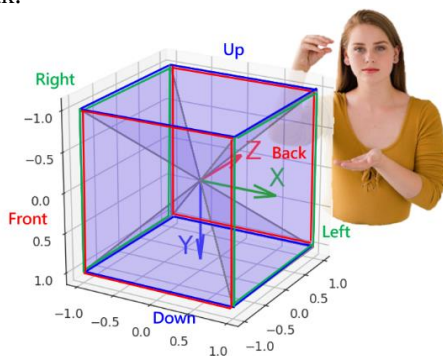


Fig. 1 Orientation types in the division of 3D space.

2.3 Location Recognition

Location recognition is to identify the relationship between the handshape and different parts of the body, including determining which area of the body the

handshape is located in and which direction it is touching or pointing to the body. We have categorized sign language positions into 23 distinct regions, such as in front of the body, above the head, face, the top of the face, forehead, eyes, nose, mouth, chin or lower face, temple, ears, cheeks, neck, shoulders, heart, chest, waist, legs, armpit, upper arm, lower arm, back of hand or back of wrist, and wrist. Since the location represents where the hand is or the body part it interacts with, we initially focus on the fingers or the palm as the points of interest based on the handshape. Subsequently, we calculate whether the distances between the key point and various region boundaries are less than a predefined threshold to determine the hand’s location. This involves computing the distances between the keypoint on the hand and reference points or reference lines. For example, to ascertain whether a finger is pointing at the forearm, we calculate the distance between the fingertip and the line segment from the shoulder to the elbow, then compare it to a certain threshold.

2.4 Non-Manual Signals Recognition

In sign language, alongside hand movements, non-manual signals like head movements and facial expressions play a crucial role in conveying emotions and meanings. These signals encompass head movements (nodding, shaking, tilting) and facial expressions, which involve movements like raising or furrowing eyebrows, wrinkling or relaxing the nose, and opening or closing the mouth. Cheek puffiness and eye openness also contribute to the nuances of sign language.

2.4.1 Head Movements

In terms of head movement recognition, we calculate the Euler angles of the head and determine the corresponding head movement based on these angles. In order to determine the head pose, we transform the pixel coordinates from the facial keypoint recognition results through the camera’s internal parameter matrix to obtain the three-dimensional coordinates of the facial keypoints in the camera coordinate system. Due to the rotational and translational relationship between the camera coordinate system and the world coordinate system, the transformation matrix between the camera coordinates and world coordinates of facial keypoints can be obtained using the Direct Linear Transform (DLT). With the rotation matrix, the Euler angles, which represent the rotational angles of the head in this context, can be derived from the relationships among the elements of the rotation matrix [3]. We define the pitch angle range to represent facing upwards or downwards, the yaw angle range to represent turning the face to the left or right, and the roll angle range to represent tilting the head to the left or right.

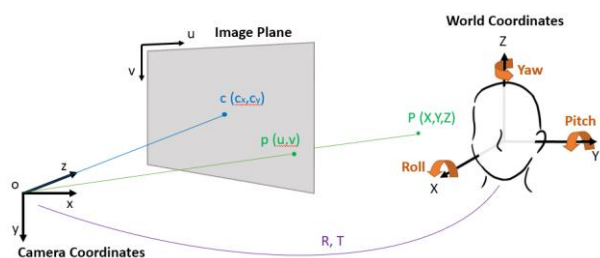


Fig. 2 Head pose estimation.

In this study, we established a hypothetical facial model to define the world coordinate system. The approach involved defining the world coordinates based on specific points on the human face, ensuring variability along all three coordinate axes. These selected points are located at the tip of the nose, the outer corner of both eyes, the midpoint between the eyebrows, and the corners of the lips (both right and left). The specific location of these selected points is shown in Fig. 3 (left).

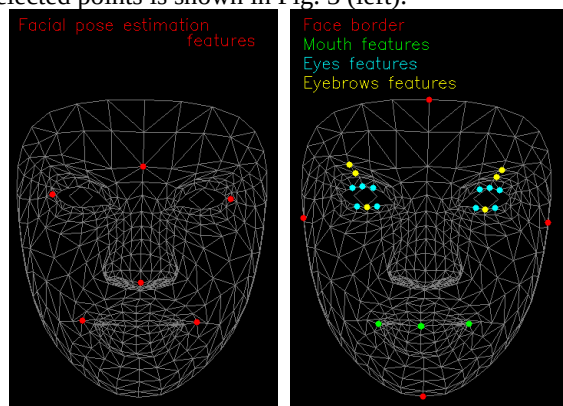


Fig. 3 (left) The selected points on the face to construct the world coordinate system. (right) Feature points related to facial features.

2.4.2 Facial Expressions

For tracking dynamic changes of the facial features, we observed the movement of face mesh keypoints where these actions will occur, and finally defined the occurrence of these actions based on the range of motion of specific selected points. The following are some common movements observed in TSL facial expressions, along with how we define their occurrence using the MediaPipe Face Mesh. Furrowing and raising the eyebrows are determined by the distance between the midpoint of the lower edge of the eyebrows and the midpoint of the upper edge of the eyes. The degree of eye openness is defined by the ratio of the average distance between the upper and lower edges of the eyes to the length of the face, allowing for categorization into open eyes, squinted eyes, and closed eyes. Whether the mouth is open or closed is determined based on the ratio of the distance between the center points of the upper and lower lips to the length of the face. Whether the mouth is grinning is determined by the ratio of the distance

between the corners of the mouth on both sides and the width of the face. The specific location of these feature points is shown in Fig. 3 (right).

2.5 ML Method for Recognition

In the realm of Sign Language Recognition (SLR), two primary approaches exist: the vision-based approach and the sensor-based approach. The vision-based approach tends to exhibit shorter time delays [4]. Moreover, TSL has more extensive use of character signing, as opposed to directly drawing a continuous complex written form of the word in midair, which is unique in the Asian Sign language sphere [5]. Hence a machine learning process is a plausible way to determine the specific vocabulary captured by a camera. ML is implemented to improve and compare with the previous direct calculation-based model in 2.1 Handshape recognition and 2.3 Location recognition. In this article, the LSTM network model is used to test the training data compared to a Recurrent Neural Network (RNN) model in terms of the impact on long-term dependencies of the sequential nature of TSL and generally sign language as a whole [6].

2.5.1 Detection and Data Library

To accomplish the aim of our study, we address the hand movement using the implementation of MediaPipe Holistic. As shown in Fig. 4, the selected points of the left hand, right hand, and face are detected and pinpointed in image coordinates, which are then transformed into the world coordinate system. To maximize the accuracy of our LSTM model, 15 frames were captured in the course of each gesture, with 10 repetitions of each TSL vocabulary to ensure data accuracy. This approach allows us to represent the spatial changes in individual signs effectively. We conducted tests using three common TSL vocabularies: “head”, “sleepy”, and “handkerchief,” all of which require the five-finger open handshape [7]. The differences in gestures lie in their positions relative to the head – at the top, in front, and beside the mouth, respectively.



Fig. 4 The selected points of the hand and face are detected and pinpointed in image coordinates.

2.5.2 Training and Testing

To process the captured data, we address the reference hand gesture signals (Fig. 5) within a single frame by comparing the similarities between the trained dataset and the testing dataset. The values of the points

are organized into separate folders for each gesture and are saved for subsequent training and testing. Our dataset comprises 33 pose landmarks, 468 face landmarks, and 21 hand landmarks per hand. Since the majority of the hand gestures require only the upper body and mainly the hand and the face, not all pose landmarks are present in the image detections. Moreover, facial changes were not particularly distinctive across the three TSL vocabularies. Therefore, the main focus of the training and testing data was on handshapes and their relative locations to the head. Due to the size of the dataset, a six-layer sequential model was employed for training the classification models, following a split of the 95% training and 5% testing split, as well as the 80% training and 20% testing split, adhering to the Pareto Principle [8]. For qualitative data assessment, we used a sequential model consisting of 3 LSTM layers and 3 dense layers. This model was trained under 200 epochs to evaluate accuracy and training loss.

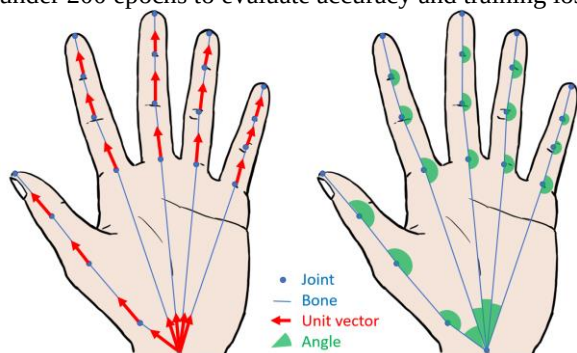


Fig. 5 The selected points on the hand, specifically the hand points, hand angles and the hand vectors.

3. Experiment Results

This section is divided into three sections. The first section presents the results of the TSL element recognition method. The second section presents the recognition outcomes of the ML algorithm. The training data for the model comes from two students and is applied to the same two individuals. In the final section, we compare and discuss the effectiveness of these two recognition methods.

3.1 Elements Recognition

The element recognition method, as described in 2.1 to 2.4, combines techniques for recognizing static TSL images. It analyzes facial expressions, head movements, handshapes, and posture in each frame to interpret the meaning of TSL gestures. This real-time method incorporates a mode filter to enhance accuracy by reducing the impact of noise or errors. However, it's important to note that it only works with static sign language gestures captured in single frames, as TSL lacks a movement element.

Fig. 6 displays the recognition results along with the skeletons for these photos. The meanings of these signs, presented from left to right and top to bottom, are as follows: “a day before yesterday”, “a weekend”, “that person”, “ears”, “the police”, “a biscuit”, “a flashlight”,

and “a heart”. In this experiment, we employed a Logitech 720p 30fps Webcam for visual recognition of TSL. The average recognition time required was 0.125 seconds. In the upper top left corner of each result image, the recognized handshape, orientation, and location for both hands are displayed. It can be observed that there are some handshape recognition errors, and in certain cases, the recognition model struggles to identify the hand after the MediaPipe Holistic model cropping the hands from the pose detection. These challenges may be attributed to the limited diversity and adequacy of the reference handshape data, as well as inaccuracies in handshape representation compared to the correct sign. Orientation of the handshape can also pose challenges for recognition. Furthermore, handshapes near the face may impact face mesh detection, causing issues with facial recognition and subsequently affecting hand position determination.

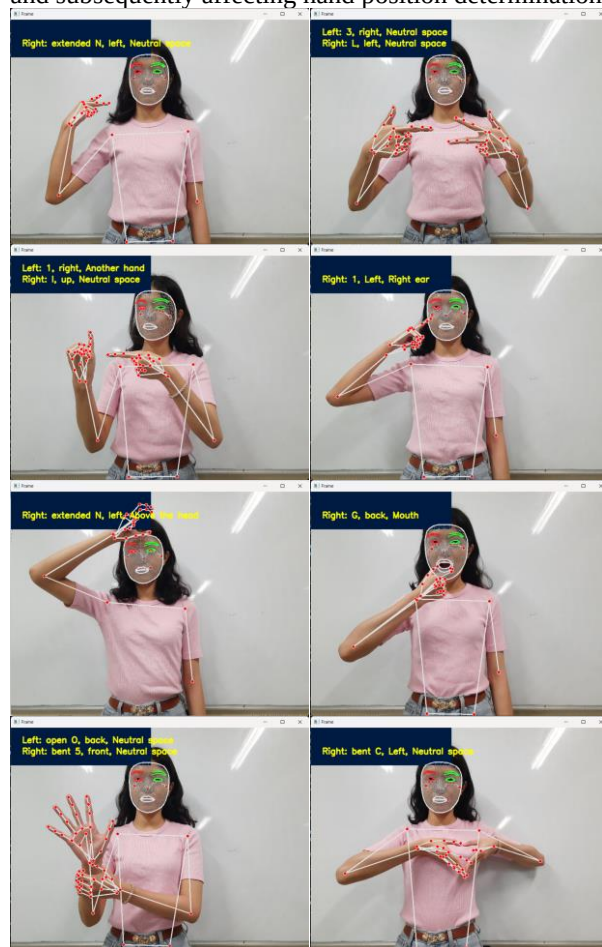


Fig. 6 The recognition results for the eight static TSL vocabulary photos.

To better illustrate the effectiveness of facial and head movement recognition, face recognition is displayed separately. Fig. 7 displays photos of recognized facial expressions and head movements, showcasing a total of six combinations of non-manual signals in TSL. In these images, deep blue text indicates various non-manual movement changes, while red text

represents the corresponding sign language meanings. One of the challenges encountered in facial expression recognition is the obstruction of facial features by hands during sign language movements. Furthermore, individuals vary in the sizes and distances between their facial features, which poses a challenge when applying a universal set of comparison standards.

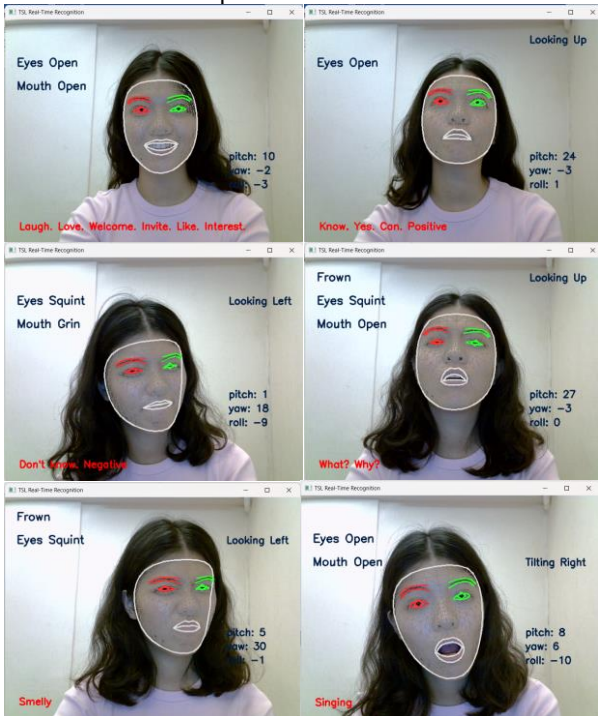


Fig. 7 The non-manual movement recognition results for the six static TSL vocabulary photos.

3.2 ML Method for Recognition

The main objective of this study is to recognize TSL through visual capture and detection. A ML approach is used in which samples of TSL gestures are captured by a built-in video camera on a laptop. The camera features a resolution of 720p without depth sensing. The research results can be categorized into two main groups based on the percentage of training and testing data:

- 1) 95% and 5% split: 28 sets of data were utilized for training, and 2 sets of data were reserved for testing. The validation accuracy stabilizes as the number of epochs approaches and exceeds 100. As for the training loss, there is a sharp increase around the 60th epochs, followed by a gradual decrease towards zero after the 80th epochs.

Table 1 The LSTM model used for the 95% and 5% split model, and the 80% and 20% split model.

Model: sequential		
Layer (type)	Output shape	Param #
lstm (LSTM)	(None, 15, 30)	203160
lstm_1 (LSTM)	(None, 15, 60)	21840
lstm_2 (LSTM)	(None, 30)	10920
dense (Dense)	(None, 30)	930

dense_1 (Dense)	(None, 15)	465
dense_2 (Dense)	(None, 3)	48
Total params: 237,363		
Trainable params: 237,363		
Non-trainable params: 0		

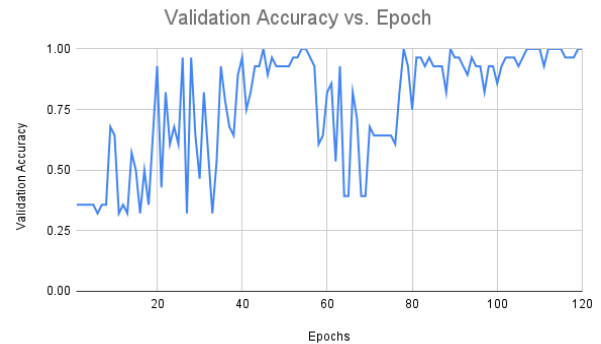


Fig. 8 The validation accuracy with epoch number line chart for the 95% and 5% split model.



Fig. 9 The training loss with epoch number line chart for the 95% and 5% split model.

- 2) 80% and 20% split: 24 sets of data were utilized for training, and 6 sets of data were used for testing. The validation accuracy stabilizes as the number of epochs approaches and exceeds 60. As for the training loss, there are sharp increases around the 20th and 40th epochs, followed by a gradual decrease toward zero after the 60th epoch.

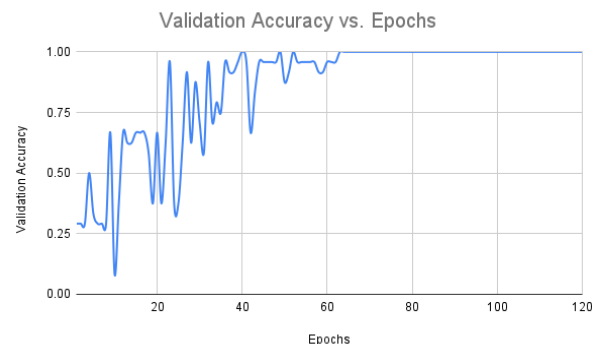


Fig. 10 The validation accuracy with epoch number line chart for the 80% and 20% split model.



Fig. 11 The training loss with epoch number line chart for the 80% and 20% split model.

3.3 Comparison and Discussion

Comparing calculation-based recognition with ML models, it becomes evident that the former is less precise, exhibits larger errors, and lacks flexibility. Calculation-based models tend to accumulate errors from various TSL element recognition results, leading to poor overall recognition performance. In contrast, ML methods can avoid this effect and do not require the disassembly and reassembly of TSL elements. Moreover, the simultaneous calculation-based recognition of TSL handshape, orientation, and location takes an average processing time of 0.125 seconds, which exceeds the desired real-time efficiency target of less than 0.1 seconds. This challenge is rooted in the inherent computational complexity of the calculation-based approach.

In our experiments, it was observed that gestures near the face could result in significant recognition errors, as well as hand occlusion problems encountered in two-dimensional images. These challenges could be better tackled through the integration of multi-vision processing to create stereo vision and higher resolution for capturing subtle facial landmarks. Lastly, it's important to note that the training and testing data for this experiment were generated by non-TSL users mimicking sign language gestures, which could potentially impact model accuracy. To develop a large and reliable model in the future, it will be necessary to have sign language users create a high-resolution TSL image dataset. Furthermore, the accuracy of the model and algorithm can also be improved by using large-scale datasets, such as the American Sign Language Alphabet Dataset.

4. Conclusions

To enable robots to understand sign language, we've developed two visual recognition methods, both relying on MediaPipe Holistic to detect key points in the hands, body, and face for recognition data. The first method focuses on separate recognition of TSL elements, including calculation-based handshape, orientation, location, and non-manual signal recognition. The second method utilizes an LSTM model to establish visual

models for TSL. It can be seen from the results that the ML method has more potential compared to calculation-based models. ML provides greater flexibility and higher recognition accuracy, avoiding error accumulation observed in multiple-element recognition. Additionally, due to the scarcity of TSL image data, building a larger-scale TSL recognition system in the future will require the development of a comprehensive TSL dataset. To solve the problem of occlusion and enhance detailed recognition, the incorporation of stereo vision with multiple camera sensors appears to be a promising direction. Lastly, in the future, the model can be further optimized by incorporating convolutional neural network (CNN) architectures to improve recognition accuracy or to enable dynamic sign language gesture recognition, achieving real-time recognition effectiveness.

5. Acknowledgement

This work was partially supported by National Science and Technology Council under grant MOST110-2221-E-002-110-MY3.

6. References

1. C. M. Chao, "Introduction: Linguistics of Sign Language," Chinese Deaf Association, Taiwan, p. 95-103, 2011.
2. Google Research, Dec. 2020, "MediaPipe Holistic - Simultaneous Face, Hand and Pose Prediction, on Device," <<https://blog.research.google/2020/12/mediapipe-holistic-simultaneous-face.html>>, Oct. 2022.
3. NASA Mission Planning and Analysis Division, "Euler Angles, Quaternions, and Transformation Matrices", NASA, 1977.
4. M. Madhwarasan and P. P. Roy, "A Comprehensive Review of Sign Language Recognition: Different Types, Modalities, and Datasets," arXiv, 2022.
5. S. Fischer and Q. Gong, "Marked Hand Configurations in Asian Sign Languages," Formational Units in Sign Languages, pp. 19-42, 2011.
6. M. Al-Qurishi, T. Khalid and R. Souissi, "Deep Learning for Sign Language Recognition: Current Techniques, Benchmarks, and Open Issues," IEEE Access, Vol. 9, 2021.
7. J. Tsay, "Taiwan Sign Language Online Dictionary: Construction and Expansion," 2nd ILAS Annual Linguistics Forum-National Language Corpora: Design and Construction, pp. 85-110, Taipei: Institute of Linguistics, Academia Sinica, 2019.
8. J. J. Gago, V. Vasco, B. Łukawski, U. Pattacini, V. Tikhonoff, J. G. Victores and C. Balaguer, "Sequence-to-Sequence Natural Language to Humanoid Robot Sign Language," arXiv, 2019.